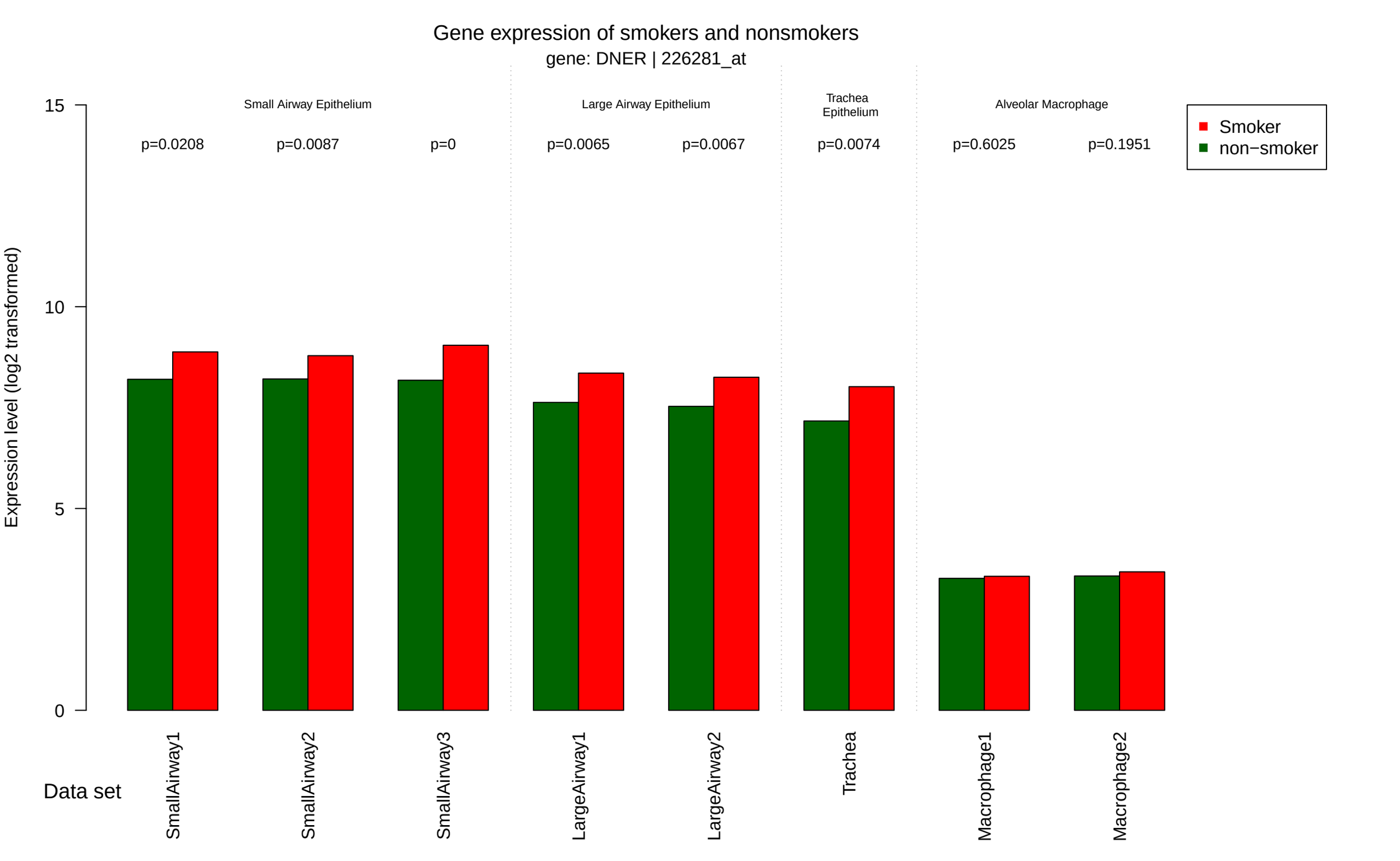
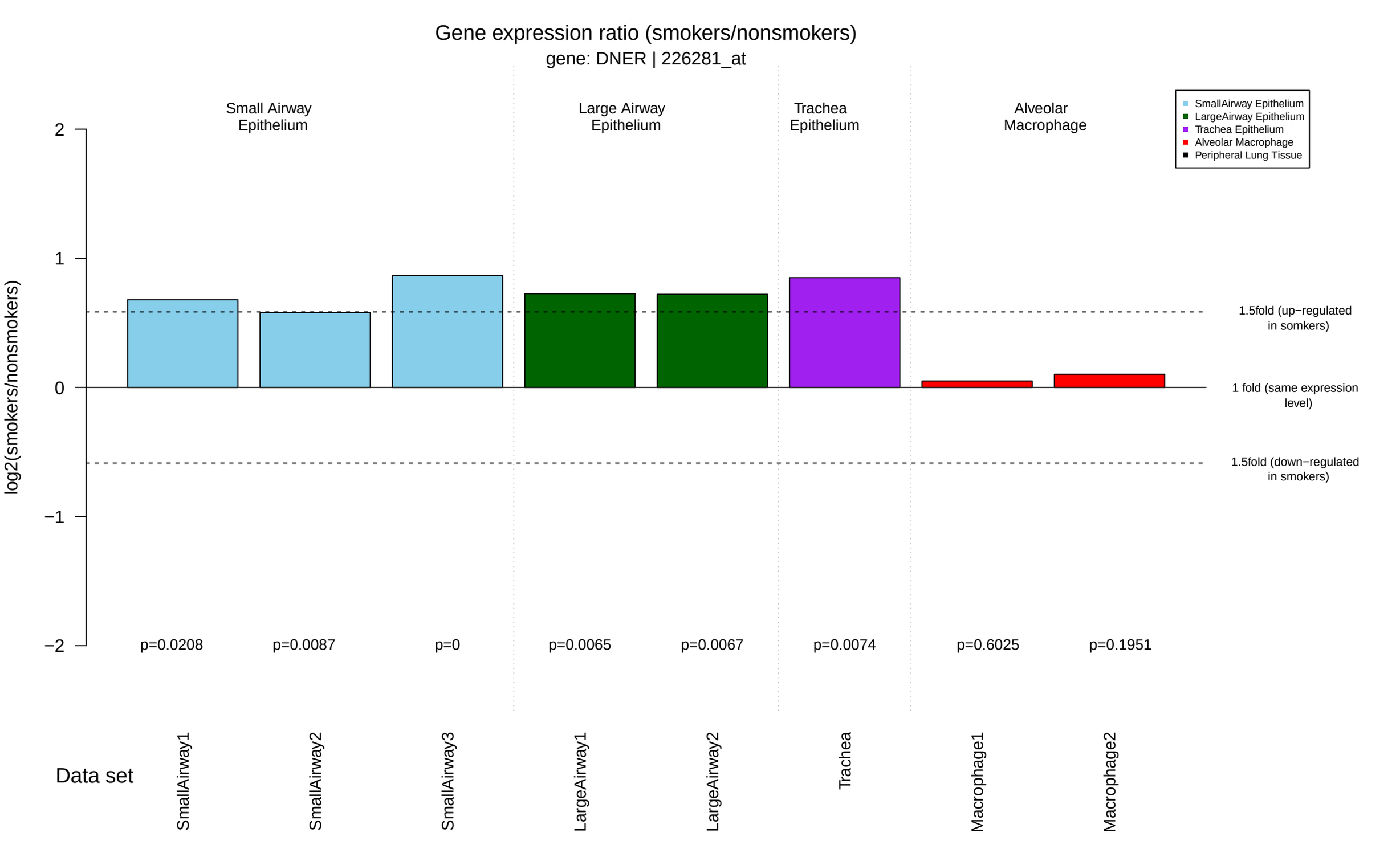
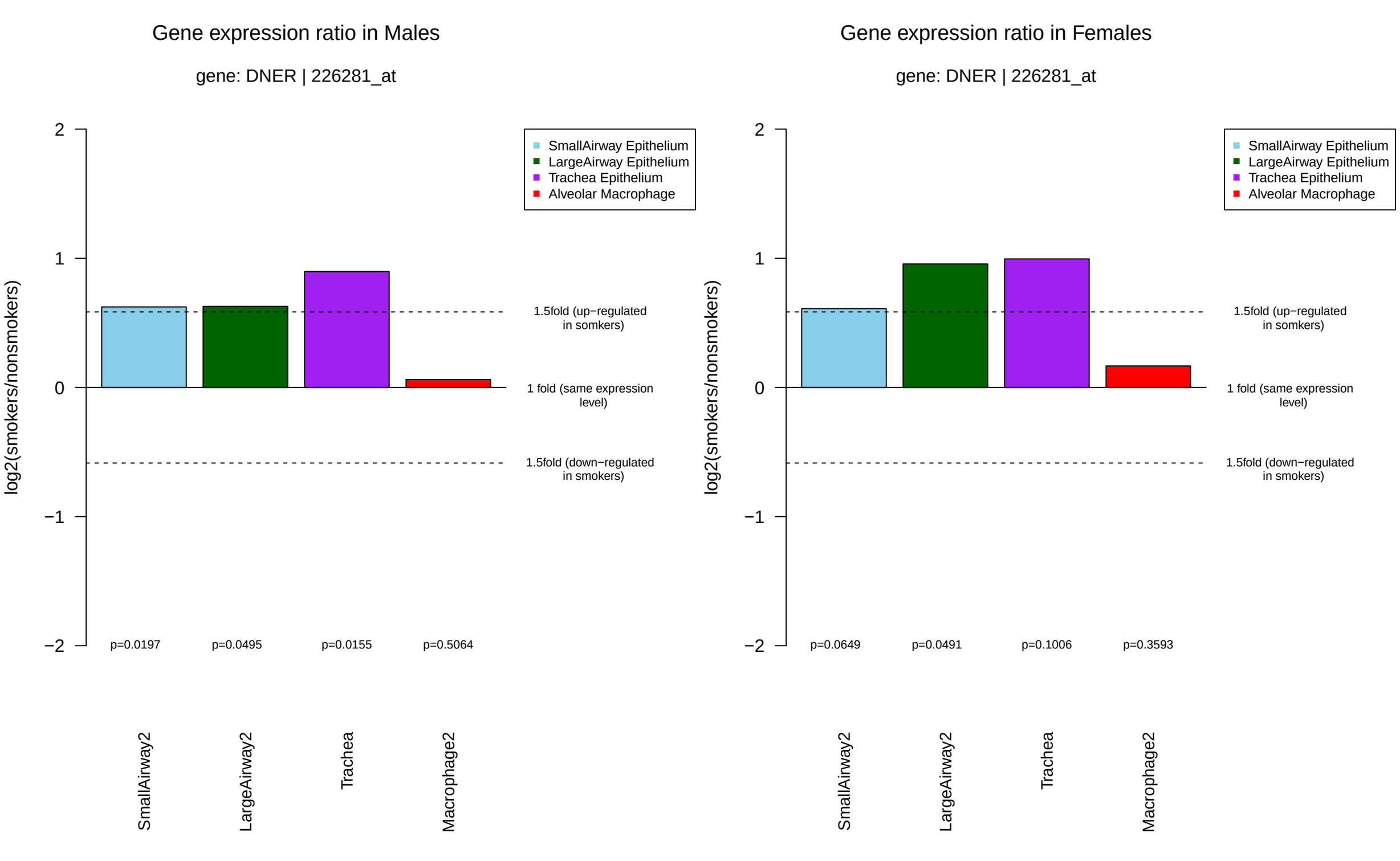
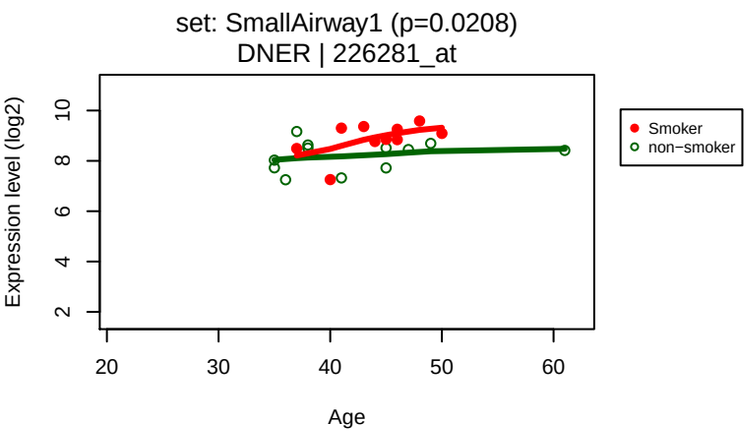
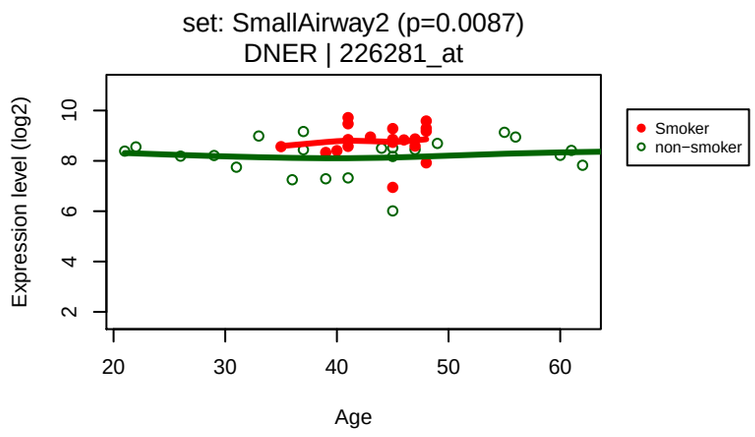
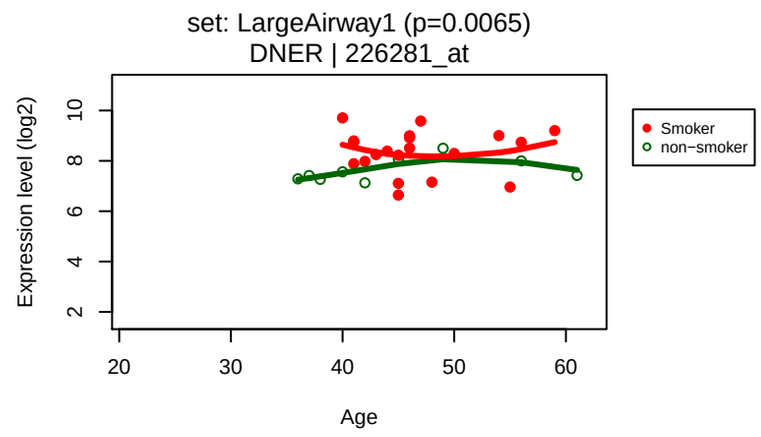

Gene expression of smokers and nonsmokers
gene: DNER | 226281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0208
p=0.0087
p=0
p=0.0065
p=0.0067
p=0.0074
p=0.6025
p=0.1951
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

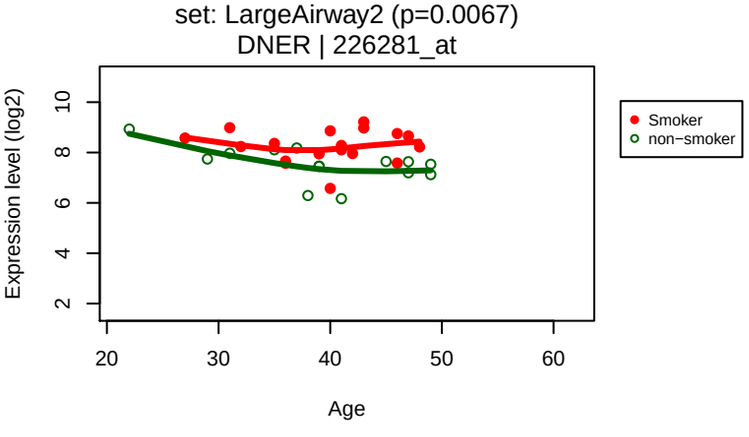

Gene expression ratio (smokers/nonsmokers)
gene: DNER | 226281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0208
p=0.0087
p=0
p=0.0065
p=0.0067
p=0.0074
p=0.6025
p=0.1951
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

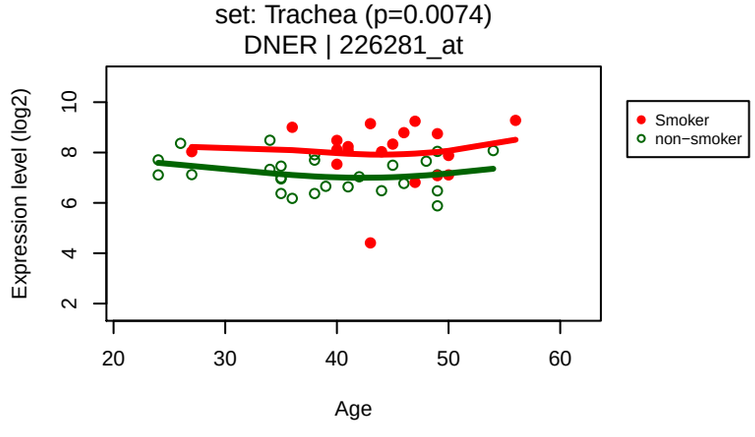

Gene expression ratio in Males
gene: DNER | 226281_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0197
p=0.0495
p=0.0155
p=0.5064
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DNER | 226281_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0649
p=0.0491
p=0.1006
p=0.3593
SmallAirway2
LargeAirway2
Trachea
Macrophage2

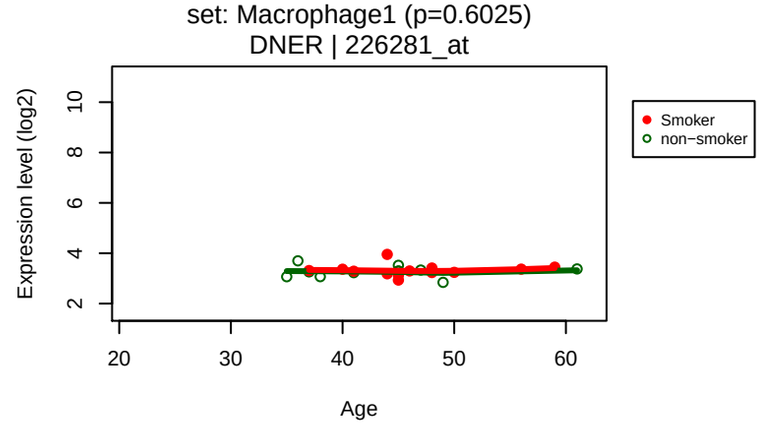

set: SmallAirway1 (p=0.0208)
DNER | 226281_at
Expression level (log2)
Age
Smoker
non−smoker

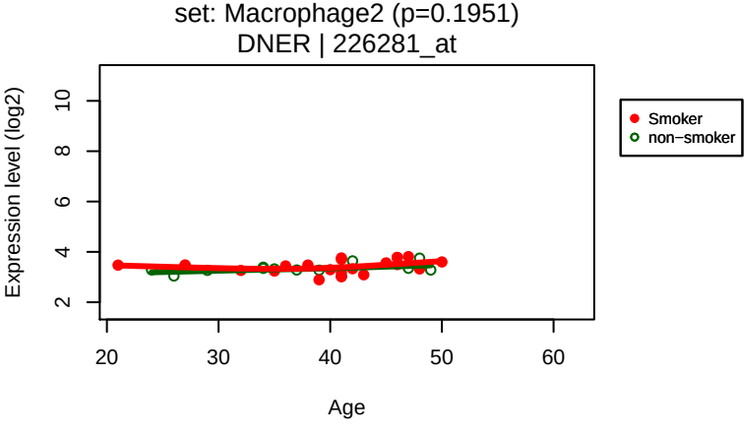

set: SmallAirway2 (p=0.0087)
DNER | 226281_at
Expression level (log2)
Age
Smoker
non−smoker

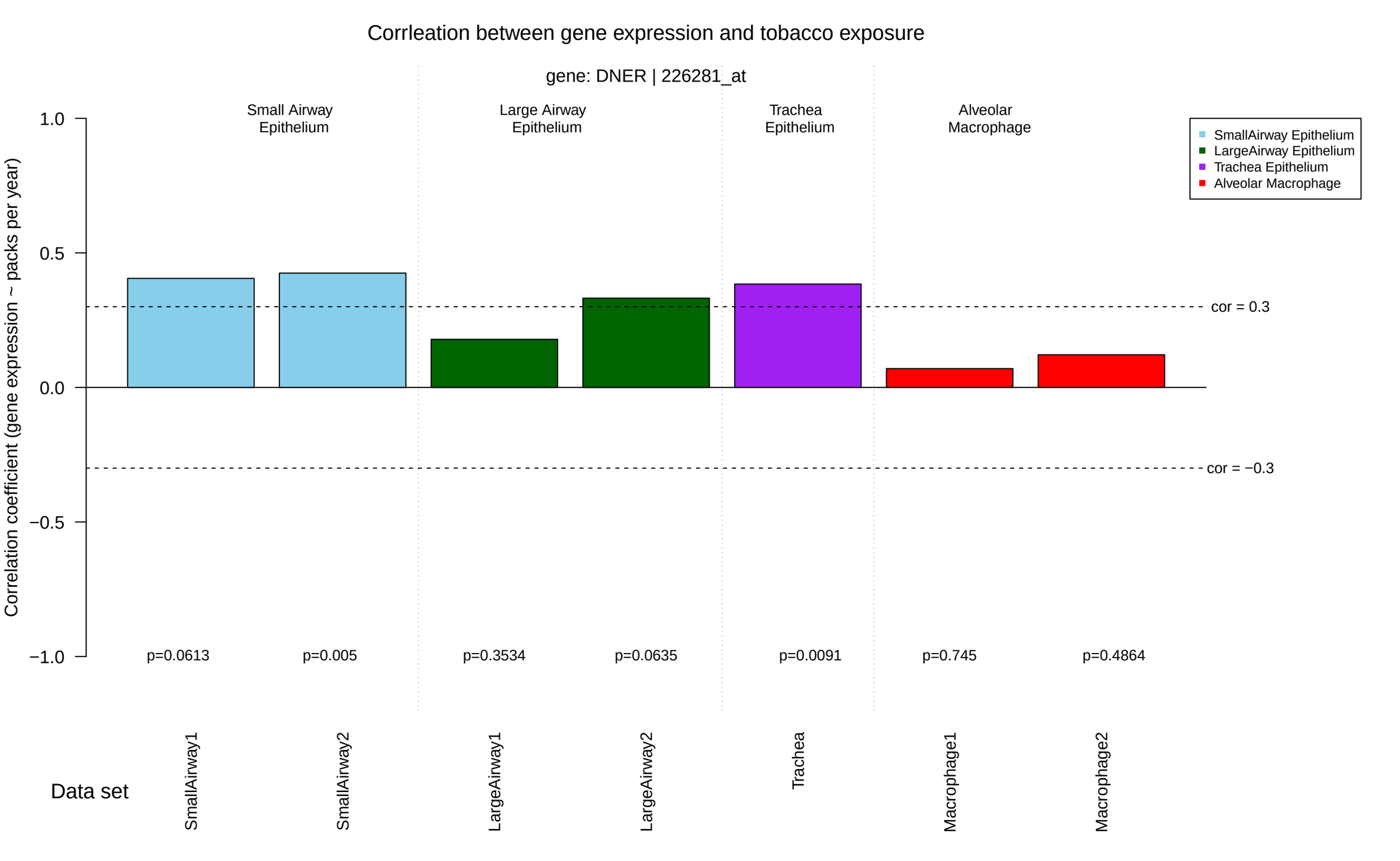

set: LargeAirway1 (p=0.0065)
DNER | 226281_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0067)
DNER | 226281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0074)
DNER | 226281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6025)
DNER | 226281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1951)
DNER | 226281_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DNER | 226281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0613
p=0.005
p=0.3534
p=0.0635
p=0.0091
p=0.745
p=0.4864
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2